

Experimental Evaluation of Near- and Far-Transfer Effects of an Adaptive Multicomponent Working Memory Training

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Summary: An important question in cognitive science today is whether working memory training leads to transfer effects. Training studies reported no transfer as well as far transfer to untrained abilities (e.g., reasoning), but different methodological problems have not allowed reliable conclusions to be drawn. Furthermore, most samples have consisted of young adults, whereas older adults have rarely been included. This study attempted to address these limitations. Ninety-one older adults were randomly assigned to a training group or an active or passive control group. Short-term memory, speed, and reasoning were assessed as far-transfer criteria. In addition, transfer to everyday life was investigated using an ambulatory assessment method. Although there were significant training effects, no transfer effects were found. Instead, the active control group showed the same increase in the near-transfer tasks (i.e., working memory) as the experimental group indicating that unspecific factors explain the training gains of older adults. Copyright © 2015 John Wiley & Sons, Ltd.

INTRODUCTION

Considering the demographic development on the one hand and the decrease of cognitive functioning during aging on the other hand, training approaches that might improve or at least maintain cognitive abilities gain more and more importance. The question therefore arises what contents the training should have. It is reasonable to focus on rather basic abilities that play a dominant role in cognitive aging and additionally are of essential importance for many higher order abilities. The development and evaluation of working memory training programs follows this strategy and has become a dominant issue (e.g., Borella, Carretti, Riboldi, & De Beni, 2010; Chein & Morrison, 2010; Jaeggi, Buschkuhl, Jonides, & Perrig, 2008; Salminen, Strobach, & Schubert, 2012).

Working memory is a limited-capacity system for temporarily storing *and* processing information while executing complex cognitive tasks (e.g., learning, reasoning, comprehension, and decision making; Baddeley, 2003, 2012; Conway, Jarrold, Kane, Miyake, & Towse, 2007; Unsworth & Engle, 2007) and is one cause of cognitive aging (e.g., Park et al., 1996; Salthouse, 1996). Further, working memory capacity is of high relevance for many higher order abilities and most areas of academic performance, for example, reasoning (e.g., Conway, Kane, & Engle, 2003; Süß, Oberauer, Wittman, Wilhelm, & Schulze, 2002), reading comprehension (e.g., Cain, Oakhill, & Bryant, 2004), language comprehension (e.g., King & Just, 1991), math performance (e.g., Swanson & Kim, 2007), and complex problem solving (e.g., Wittmann & Süß, 1999) as well as for a broad spectrum of applications (e.g., understanding of clinical disorders, neurocognitive interventions, and language therapy).

Multicomponent models of working memory assume several content-specific storage systems (e.g., phonological loop and visuospatial sketchpad) and a control unit (i.e., the cen-

tral executive; Baddeley & Hitch, 1974; Baddeley, 2012). Basic functions of the latter are to focus or divide attention, to switch between tasks, and to inhibit information from working memory that is not relevant or no longer relevant (i.e., basic executive functions; Baddeley 1996; Engle & Kane, 2004; Miyake, Friedman, Emerson, Witzki, Howerter, & Wager, 2000). Based on these components, it is possible to make assumptions concerning the constructs where transfer effects may be expected after working memory training, and this helps to determine the corresponding distance of those transfer effects. More detailed, transfer indicates the expected carry-over effect to untrained constructs (or tasks, respectively; Barnett & Ceci, 2002) via overlaps of involved processes (von Bastian & Oberauer, 2013a). The term 'near' transfer means a transfer task indicating the same construct as the training task (e.g., two different working memory capacity tasks), 'moderate' describes transfer to tasks of constructs that share a major part of variance with the trained construct (e.g., working memory capacity and short-term memory), and 'far' means transfer to tasks tapping resources in a related but disparate cognitive domain, that is, indicating strongly related but different cognitive constructs (e.g., working memory capacity and reasoning). Figure 1 illustrates these transfer distances.

In a narrow sense, the distance of transfer depends on the relation of trained construct and transfer construct: The higher their correlation is, the more likely and higher are the expected transfer effects. In a broader sense, transfer effects are aiming at the utility of the training program investigating, which cognitive abilities and cognitive performances in daily life profit from training gains and how strong the effects are. Obviously, transfer effects can only occur if training is effective at all and if trained functions are meaningful for the transfer construct. The more meaningful the trained functions are—that is, the more working memory capacity is involved in the transfer construct—the higher and more likely are transfer effects. Further, it is often assumed that near transfer is a requirement for far transfer; thus, working memory capacity has to be improved at the construct level before transfer to reasoning can occur (Harrison et al., 2013).

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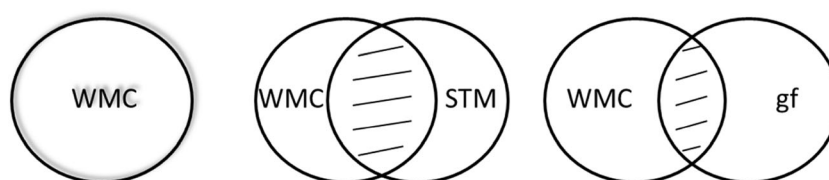


Figure 1. Schematic representation of transfer distances. Near transfer: Two tasks tapping the same construct (e.g., WMC) show almost complete overlap with only task-specific differences (e.g., different contents), so transfer is very likely. Moderate transfer: Two tasks of closely related constructs (e.g., WMC and STM) share a great amount of their variances but there are still unique parts; the likelihood of transfer is lower here. Far transfer: Indicated constructs of two tasks show overlaps (e.g., similar task requirements, characteristics, limitations, or restrictions) and a great amount of unique variance (e.g., WMC and gf). The bigger the unique parts are, the more unlikely is transfer.

WMC = working memory capacity; STM = short-term memory; gf = reasoning

Review of working memory training studies

Working memory training studies can be roughly classified into four categories depending on whether or not the training was effective and whether the sample consisted of younger or older adults. Earlier training studies investigated transfer effects in younger adults and, to a certain extent, reported a considerable amount of far transfer to reasoning (e.g., Jaeggi et al., 2008). On this basis, subsequent studies tried to replicate and specify these promising results, but there were an increasing number of studies that reported zero effects instead (Chooi & Thompson, 2012; Harrison et al., 2013; Owen et al., 2010; Redick et al., 2013; Thompson et al., 2013). With only a few exceptions (e.g., Stepankova et al., 2013; von Bastian & Oberauer, 2013b; Zinke et al., 2013), studies with older adults have typically reported smaller (compared with younger adults) or nonexistent transfer effects (e.g., Borella et al., 2013a; Brehmer, Westerberg, & Bäckman, 2012; Buschkuhl et al., 2008; Dahlin, Nyberg, Bäckman, & Neely, 2008; Heinzl et al., 2014; Li et al., 2008; Schmiedek, Lövdén, & Lindenberger, 2010; Xin, Lai, Li, & Maes, 2014; Zinke, Zeintl, Eschen, Herzog, & Kliegel, 2012). It seems that an *effective* training in younger adults *does not* necessarily produce transfer effects in older adults and an *ineffective* training in younger adults *cannot* produce transfer in older adults. On the other hand, an *effective* training in older adults should also be effective in younger adults with even greater results. Thus, if a specific training is *ineffective* in older adults, it is uncertain whether particularities of the aging population are responsible (see next section about challenges), and this training would be effective in younger adults, or if the training program was not effective at all and no transfer effects could be expected.

Different training approaches used different training tasks (e.g., dual n-back, complex span tasks, and updating tasks), transfer tasks (i.e., different constructs were assessed as well as different reasoning tasks were used [but mostly Raven Advanced Progressive Matrices test; Raven, 1990]), length of training (ranging from 3 sessions to 100), or time intervals between sessions (ranging from daily to weekly sessions) so that factors influencing transfer could be detected, for example, by meta-analyses and systematic reviews. However, results show a high variability. It is therefore difficult to summarize them and to draw consistent conclusions. As participants' age in particular seems to play a major role, meta-analyses and reviews were separately conducted for young adults (e.g., Au et al., 2014; Klingberg, 2010;

Melby-Lervåg & Hulme, 2012; Morrison & Chein, 2011; Shipstead, Redick, & Engle, 2012a) and old adults (e.g., Borella, Carretti, Zanolini, Zavagnin, & De Beni, 2013b; Hindin & Zelinski, 2012; Lustig, Shah, Seidler, & Reuter-Lorenz, 2009; for both age groups: von Bastian & Oberauer, 2013a). Based on these reviews, we will recapitulate the challenges in the field of working memory training in the next two sections, starting with general challenges and continuing with specific challenges of implementing a working memory training study in an aging cohort.

Challenges for the field of working memory training

Four main aspects might distort the reported results. The first two issues rather overestimate transfer effects; the latter two underestimate them. (i) Studies reporting far transfer have frequently used single-outcome tasks (e.g., Raven Advanced Progressive Matrices test) as indices of improvement in an entire cognitive domain (e.g., reasoning), but some task-specific aspect of function has probably changed rather than improvement of the entire cognitive operation. (ii) Another limitation has been the common use of only a passive no-contact control group without an active control group, but only the latter can control for placebo and expectancy effects. By contrast, some studies reporting only near transfer have not offered adequate transfer conditions (Morrison & Chein, 2011; Shipstead et al., 2012a): That is, the training was (iii) nonadaptive or (iv) not diversified. Adaptiveness is important to ensure an adequate level of difficulty that does not overburden or underchallenge participants (optimal stimulus and motivational impact; but see von Bastian & Oberauer, 2013a, for a probably minor role of adaptivity in the effectiveness of training). Thus, participants work continuously at their individual performance limit (see Vygotski, 1987, *zone of proximal development*). If the training is not diversified (i.e., only one training task was used), the development of task-specific strategies is possible without enhancing the intended ability construct. Single-task training may also lead to boredom. On the contrary, a variable training with different tasks prevents this kind of method variance, and it allows an examination at construct level instead of only at task level. Nevertheless, a variable training does not unambiguously allow attribution of near transfer, that is, transfer from a specific task to another task indicating the same construct, because it is not clear which task(s) contribute to any specific instance of transfer or whether all tasks are equally efficacious in terms of producing transfer effects.

Furthermore, participants spend reduced time training with each task in a variable training if complete training time is kept constant.

Apart from the previously described issues of a potentially successful training, studies with an aging cohort include additional challenges. The expected training gains and transfer effects of older adults are smaller than the expected gains of younger adults because of the life span development of fluid intelligence (Baltes, 1987; Horn & Noll, 1997; cf. Baddeley, 2012) and the reduced neural plasticity of older adults. However, studies using the testing-the-limits paradigm also show that cognitive plasticity can be assumed even in old age, although it is constricted in comparison with that of young adults (e.g., Kliegl, Smith, & Baltes, 1989; Lindenberger, Kliegl, & Baltes 1993). Thus, cognitive training in older adults can be successful because there is still room for improvement (e.g., Strobach, Frensch, Müller, & Schubert, 2012a) and older people may have higher motivation owing to increased meaningfulness facing age-dependent changes.

However, difficulty of tasks, familiarity with tasks, amount of required instruction and examples, usage of new technology, and resilience during training sessions might be specifics that distinguish training with older adults from training with younger adults. Furthermore, transfer to everyday life gain more importance in order to maintain or improve functionality and independence in old age, but those investigations have rarely been included in training studies. One exception is the study by Richmond, Morrison, Chein, and Olson (2011) who used the performance-based Test of Everyday Attention (Robertson, Ward, Ridgeway, & Nimmo-Smith, 1994) to examine transfer to everyday life. These authors applied Chein and Morrison's (2010) training paradigm (20 sessions with one verbal and one spatial working memory span task; 15 minutes for each task) to 21 older adults with an age range of 60 to 80 years. An active control group ($N=19$) completed quizzes. There was near transfer to another verbal working memory task, far transfer to a list-learning task, but no far transfer to reasoning or everyday life. However, their use of a performance-based everyday life criterion has to be emphasized because if training were to at least lead to the support and enhancement of everyday life functioning in elderly people, it would be of special importance—even without far-transfer effects measured in the lab. The training group self-reported improvements in attention, but the results of the TEA did not show these changes owing to a ceiling effect at pretest—a frequent problem of everyday life tasks. The rather low sensitivity and difficulty often prevents a measurable change for healthy older people. Additionally, transfer to everyday life is always a transfer from lab to field, too, so even if there might be a carry-over effect, it might be very difficult to statistically detect this effect.

Specific hypothesis and objectives

Despite these challenges, we investigated training effects on older adults because of the higher practical relevance of supporting the elderly, if possible, even in their daily lives. Our intention was not to show the highest possible effects but to find a useful way to help the elderly coping with their

daily routine. As there is still ambiguity with respect to transfer effects of working memory training in older adults, this study aimed to provide promising conditions by which to promote transfer as proposed by Morrison and Chein (2011) as well as by Shipstead et al., (2012a) as described in the Introduction section. Thus, training success as well as transfer criteria would need to be assessed on more than one task (multiple indicators of every transfer construct), the study should include a passive and active control group, and the training would need to be adaptive (task difficulty would need to change with performance) and variable (multicomponent training with different training tasks: complex span, updating, and switching tasks).

The intervention was based on the faceted model of working memory by Oberauer, Süß, Wilhelm, & Sander (2007): According to this model, working memory functions can be classified along two dimensions (i.e., facets) distinguishing between *contents* (verbal-numerical and visuospatial) and *cognitive operations* (simultaneous storage and processing, coordination [i.e., relational integration], and supervision [i.e., executive functions such as switching ability]). As storage and processing and coordination could not be separated (Oberauer, Süß, Schulze, Wilhelm, & Wittmann, 2000), we substituted the functional category coordination with updating. Thus, we used a multicomponent working memory training that included every possible combination of contents and cognitive operations.

Four kinds of transfer distances were examined: (i) *near* transfer to nontrained tasks involving the same construct (working memory and switching), (ii) *moderate* transfer to tasks representing a different but closely related construct (short-term memory), (iii) *far* transfer to a different construct (reasoning and processing speed), and (iv) transfer to everyday life (cognitive failures) with an innovative assessment method. In the following, we specify and justify our hypotheses: (i) We expected improvements in working memory capacity and switching ability independent of the specific tasks or contents; this near transfer has already been shown in many other studies. (ii) Short-term memory and working memory share a storage process; thus, tasks involving these two constructs contain the recall of previously presented stimuli. (iii) The common restrictions of reasoning and working memory capacity are the number of simultaneously maintained information units (Fukuda, Vogel, Mayr, & Awh, 2010) as well as the ability to focus, divide, and redirect one's attention (Oberauer et al., 2007; Oberauer, Süß, Wilhelm, & Wittmann, 2008). Therefore, a transfer effect after working memory training was expected, but there are also parts that working memory and reasoning do not share (e.g., knowledge and task-specific strategy demands; Wittmann, Süß, Oberauer, Schulze, & Wilhelm, 1995). However, assuming only direct effects of training (i.e., based on similar mechanisms), near transfer is a requirement for (moderate and) far transfer (Harrison et al., 2013; Shipstead et al., 2012a). Thus, transfer to short-term memory, speed, and reasoning was only expected after near transfer was found. In advancing age, most cognitive abilities and deficits are related to mental speed (Salthouse, 1996). Among other aspects, working memory capacity depends on the speed of rehearsal processes because the longer this inner speech takes, the more likely there will be a loss of the memory trace (Barrouillet,

Bernardin, & Camos, 2004). An improvement in working memory and especially in switching might therefore lead to a higher processing speed. (iv) The main causes of cognitive failures are supposed to be distraction, other slips of attentional control, and memory deficits (e.g., Efklides & Sideridis, 2009; Martin & Jones, 1983). In particular, mind wandering is more frequently experienced by people with a lower working memory capacity, at least in certain situations (Kane et al., 2007; McVay, Kane, & Kwapil, 2009). Thus, we expected that working memory training would decrease the number of cognitive failures.

METHOD

Participants

Participants were recruited via newspaper advertisements and flyers and from senior studies as well as sport groups. Participation was voluntary. A total of 106 older adults initially began the study, and 91 completed it; participants' reasons for dropping out were mostly due to illness or time conflicts. The sample consisted of 41 men and 50 women with a mean age of 67.97 ($SD=4.04$, range: 60–76 years) and 13.99 years of education ($SD=2.54$). Participants had normal or corrected-to-normal vision and hearing and reported good to satisfactory subjective health. Exclusion criteria were seriously neurological diseases, a cognitive dysfunction such as mild cognitive impairment or Alzheimer's disease, and the use of drugs that influence the cognitive abilities. These criteria were clarified via telephone conversation; standardized screening instruments were not used. Eighty-four participants were already retired, five were unemployed, and two participants were employed.

Pretest and posttest assessment

Based on the different transfer expectations mentioned earlier, six constructs needed to be examined: trained and untrained working memory capacity, switching ability, short-term memory, processing speed, and reasoning. Every criterion was assessed by at least three tasks with different contents (verbal, numerical, and figural–spatial). Thus, our test battery was large enough to demonstrate improvements beyond single-task learning, but it was small enough to prevent cognitive depletion or fatigue (Green, Strobach, & Schubert, 2013).

For working memory capacity, complex span tasks (i.e., there was a simple processing task between presentation and recall of stimuli) and updating tasks (i.e., stimuli had to be manipulated before recall) were administered: reading span (processing task: decision about the plausibility of short sentences; stimuli to memorize: words; Daneman & Carpenter, 1980), operation span (OS; processing task: decision about the correctness of simple equations; stimuli to memorize: words; Turner & Engle, 1989), dot span (DS; processing task: decision about the symmetry of dot patterns; stimuli to memorize: dots; Oberauer, Süß, Wilhelm, & Wittman, 2003), memory updating numerical (MUN; stimuli to update: numbers in a 3×3 grid; Oberauer et al., 2003), swaps (stimuli to update: the order of three letters; Stankov, 2001),

numerical running memory span (stimuli to update: numbers; Pollack, Johnson, & Knaff, 1959), and figural running memory span (fRMS; stimuli to update: the position of colored fields in a 3×3 grid; Pollack et al., 1959). Every stimulus was presented for 2000 milliseconds. Although many authors have used the n-back task as an indicator of working memory (e.g., Chooi & Thompson, 2012; Jaeggi et al., 2008), we—as well as Xin et al., (2014)—chose running memory span tasks because the n-back task requires a recognition of stimuli, whereas running memory span tasks, along with every other working memory task, require a recall of stimuli (Shelton, Metzger, & Elliott, 2007). In addition, the n-back task is very difficult and probably particularly unfamiliar to older adults. There were three trials of each level of difficulty, and all tasks began with two items and ended with five to six items. We alternated related constructs and contents so that two consecutive tasks were as different as possible. The tasks for pretesting and posttesting and their order were identical.

Typical switching tasks consist of two different task sets, and participants have to switch from one set of instructions or question to the other. Switching costs, the difference between reaction times on single blocks (only one task set) and mixed blocks (both task sets) was assessed with three different tasks, and the stimuli had to be evaluated according to different instructions or questions: verbal memory set switching (Question 1: Is the present word one of the two initially memorized animals?; Question 2: Is the present word one of the two initially memorized plants?), numerical response set switching (Question 1: Is the present number odd or even?; Question 2: Is the present number even or odd?), and figural task set switching (Question 1: Is the present arrow at the bottom or at the top?; Question 2: Does the present arrow point upward or downward?; Oberauer et al., 2003; Sander, 2005). Participants should answer these questions with two keys, left and right. At the numerical response set switching task, for example, they had to press the left key if the presented number was odd and they had to press the right key if the number was even. After the switch cue (the presented number turned unpredictably from white to red), they had to mentally switch the response keys, so an odd number required the right key and an even number the left key. These tasks consisted of 100 trials in four single blocks (only one question) and 300 trials in 15 mixed blocks (both questions from one switching task).

A further task was newly developed, the *multiple switching task* (MST), which required participants to switch between two to five questions (letter vocal-consonant; number odd–even; letter black–green; number lower–higher than 5; letter straight–tilted). The switching cue was either external or internal. Every possible question was listed at the top of the screen, and a red bar showed the actual question. As soon as it moved to another question, participants had to change their response behavior adequately (external condition). During the internal condition, questions were also asked, but instead of the red bar, there was a 10-second countdown at the bottom of the screen. As soon as it reached zero, participants had to change their response behavior (i.e., move to the next question) until the next countdown reached zero (Figure 2). One block of either switching cue

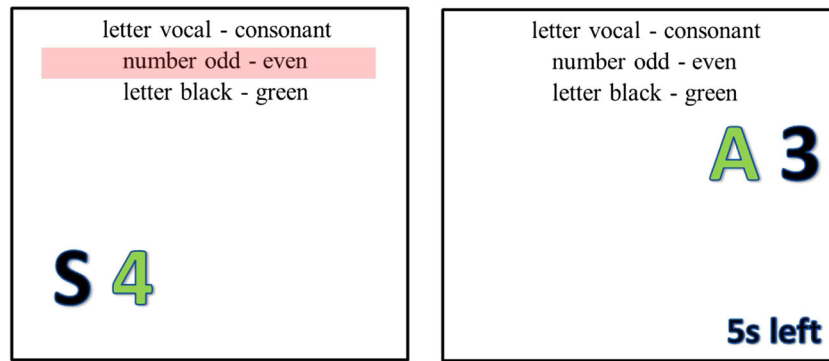


Figure 2. Multiple switching task. Participants had to press the left key if the present number–letter pair was consistent with the left word (i.e., vocal, odd, and black), and they had to press the right key if the right standing word applied (i.e., consonant, even, and green). The switching cue was either external with a red bar moving from one instruction to another (left part of this figure) or internal with a 10-second countdown (right part of this figure)

consisted of 30 letter–number pairs. The countdown condition with more than two questions is very challenging, and, therefore, improving this ability should definitely lead to transfer to other switching tasks (i.e., near transfer), and it also has a high potential to create far transfer because of its high cognitive demand.

Short-term memory was assessed by simple span tasks that were also computerized. Participants had to recall two to seven verbal (simple word span), numerical (simple digit span), or figural–spatial items (simple DS) that were presented consecutively.

Processing speed and reasoning were measured by paper-pencil tasks from the Berlin Intelligence Structure Test (Jäger, Süß & Beauducel, 1997; see Süß et al., 2002, for an English description), each with three different content areas: classification (verbal processing speed), divide by seven (numerical processing speed), old English (figural processing speed), word analogies (verbal processing capacity), matrices of numbers (numerical processing capacity), and analogies (figural processing capacity). See Table 1 for a classification of every task along the content and ability dimensions.

Two self-reports with different methodologies were used to assess transfer to everyday life. The paper-pencil based *Cognitive Failures Questionnaire* (CFQ; Broadbent, Cooper, Fitzgerald, & Parkes, 1982) was applied in the German version by Klumb (1995) with 32 items (answers *never* to *almost always*), and participants had to consider the last 6 months in

their responses. Although the CFQ has often been criticized (cf. Wilhelm, Witthöft, & Schipolowski, 2010), and it might assess training-related expectancy effects rather than cognitive failures because it is almost impossible to remember exactly and aggregate specific slips and lapses—even after a few days, we used it for reasons of comparability. The second measure was a newly developed and innovative instrument called the *electronic Questionnaire for Cognitive Failures in Everyday Life* (eKFA; Lange & Süß, 2014). Via ambulatory assessment, 13 questions (e.g., ‘Did you misplace something?’ or ‘Did you forget to say or do something?’) were presented four times a day on mobile phones; participants had to answer with absolute frequencies (answers *no*, *yes once*, *yes twice*, and *yes several times*) with respect to the previous 2 hours.

The pretest began with a biographical questionnaire. On the posttest, we applied the Big Five Inventory (Rammstedt & John, 2005). Finally, participants evaluated their participation by evaluating feelings of joy, demands, and subjective changes during the study.

Training

The training was adaptive, variable, and manifold, applying five tasks that differed in content as well as in required cognitive processes in a random order (i.e., no training session equaled another). The tasks were OS, DS, MUN, fRMS, and the MST (see previous chapter for descriptions). Thus,

Table 1. Classification of tasks along dimensions content and ability

	Verbal	Numerical	Figural–spatial
Working memory capacity			
Complex span tasks	Reading span	OPERATION SPAN	DOT SPAN
Updating tasks	Swaps	MEMORY UPDATING	FIGURAL RUNNING
		NUMERICAL	MEMORY SPAN
Switching	Memory set switching	Numerical running memory span	
	MULTIPLE SWITCHING TASK ^a	Response set switching	Task set switching
Short-term memory	Simple word span	Simple digit span	Simple dot span
Processing speed	Classification ^b	Divide by seven ^b	Old English ^b
Reasoning	Word analogies ^b	Matrices of numbers ^b	Analogies ^b

Note: Training tasks are capitalized.

^aMultiple switching task includes verbal, numerical, and figural–spatial elements.

^bPaper-pencil tasks from the Berlin Intelligence Structure Test.

Table 2. Demographic characteristics of experimental group, active control group, and passive control group

	EG	aCG	pCG
<i>N</i>	31	31	29
Gender	14 male, 17 female	15 male, 16 female	12 male, 17 female
Age	$M = 66.85$ ($SD = 3.99$)	$M = 68.23$ ($SD = 3.64$)	$M = 68.69$ ($SD = 4.14$)
Years of education	$M = 14.03$ ($SD = 2.40$)	$M = 14.42$ ($SD = 2.81$)	$M = 13.48$ ($SD = 2.40$)

Note: EG = experimental group; aCG = active control group; pCG = passive control group.

two complex span tasks, two memory updating tasks, and one switching task were applied. The adaptive algorithm of the four working memory capacity tasks led to an increase in difficulty (i.e., the number of items to be recalled) if at least three of the four trials were correct, and it led to a decrease if none or one of the four trials was correct. A trial was counted as correct if more than 80% of the items could be recalled. The first four training sessions started with two stimuli to recall, and Sessions 5–12 started with three stimuli to recall. The number of examples per task was also reduced from three to one after Session 4. The highest number of stimuli to recall depended on participant's performance but the highest possible number was nine. Two tasks ended according to a performance-based stop criterion (MUN and DS), and the other two tasks ended after a certain time (10 minutes; fRMS and OS). In the MST, the number of task sets increased if at least 27 out of 30 decisions in both switching cue blocks (external and internal) were correct, and it decreased if fewer than 27 decisions in the external switching cue block were correct. This task started with two task sets, and the maximum was seven. Training time was 10 minutes from Session 1 to Session 4 and was extended to 15 minutes in Session 5 to Session 12. Breaks between training tasks were self-paced.

The adaptive algorithm of every task did not carry over training gains from one session to the next, so participants started every session with the lowest difficulty in order to have some warmup. However, we increased this starting level and decreased the number of examples after Session 4 in order to increase the effective training time.

Design and procedure

Participants were randomly assigned to the training group or to one of the two control groups (active or passive). The investigator explained to every group that the intervention was expected to produce cognitive improvements. The training group, active control group, and passive control group did not differ significantly in age, education, and other demographic characteristics such as gender ratio and subjective health (Table 2).

Every group completed 2 days of pretesting and 2 days of posttesting (3 hours each day with one break) and 1 week of ambulatory assessment (eKFA). The study design is shown in Figure 3. Every task was explained in detail, and up to three example tasks were always given. Participants were introduced to the use and handling of the mobile phones and examples of the cognitive failures that would be asked about at the end of the second test session. The training group received two sessions of approximately 60 minutes/week (6 weeks with two sessions of 60 minutes each = 12 hours of

training). We chose this time frame because longer intervals between training sessions can lead to higher training effects than shorter intervals or even daily sessions (Penner et al., 2012). The active control group spent the same amount of time taking nonadaptive computer based quizzes (multiple choice questions about general knowledge) and playing simple board games like Ludo (6 hours each for a total of 12 hours). The passive control group was not contacted during the training phase. Posttest assessment followed on average 1 week after the last training session. Data collection took 9 months. Testing and training were carried out in groups of three to 10 persons in the lab.

Data analysis

For every span task, standardized scoring with partial credit was used, that is, scores contained the level of difficulty achieved and the proportion of trials solved on the next level. For example, there were three items to recall at Level 3; if a participant was able to recall these three items but remembered only two out of four items at Level 4, the person's achieved score would be 3.5 (50% of the Level 4 items were solved correctly). This offers an adequate combination of exhaustive information use (i.e., obtaining a large amount of information from the task) and informative value (i.e., the results can be easily interpreted; Conway et al., 2005). Switching tasks were analyzed by calculating general switching costs (the reaction time on mixed blocks was subtracted from the reaction time on single blocks) and the switching ratio (quotient of reaction time and accuracy that combines the two switching task performance indices). Specific switching costs were not considered because research has shown that training has less influence on them (Minear & Shah, 2008; Strobach, Liepelt, Schubert, & Kiesel, 2012c) and they are less related to age (Kray & Lindenberger, 2000). Reaction times lower than 200 milliseconds were set to 200 milliseconds (about 0.2%), reaction times higher than 3 *SDs* above the mean of the corresponding group were set equal to 3 *SDs* (about 11%), and the mean was substituted for trials that were answered incorrectly (about 12%). Outliers were not rejected in order to maintain a comparable number of reaction times for every task and every participant.

For each construct, an exploratory factor analysis was performed with all tasks that pertained to the one construct in order to eliminate task-specific variance.¹ Factor scores based on the first unrotated factor (i.e., the factor that explained most variance before rotation) were used for the

¹ Latent analyses were not performed because the sample size was too small (Wolf, Harrington, Clark, & Miller, 2013).

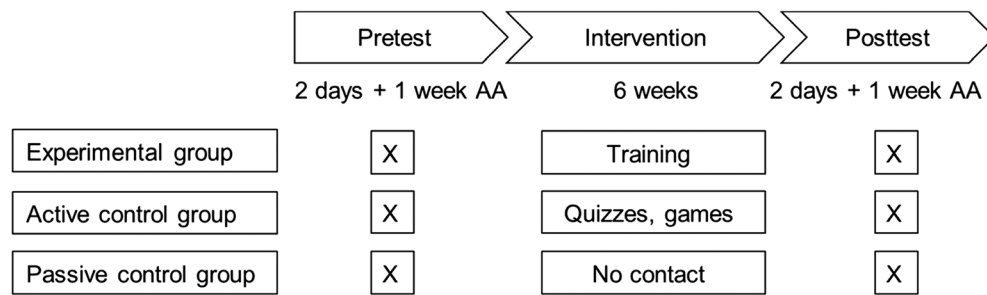


Figure 3. Flow diagram of study design. Every group completed 2 days of pretest, 2 days of posttest, and additional 1 week of ambulatory assessment each. Pretest and posttest were identical with the exception of a biographical questionnaire at pretest and the Big Five Inventory as well as an evaluation questionnaire at posttest. Groups differed in their kind of 6-week intervention. AA = ambulatory assessment

following 2 (pretest, posttest) $\times$ 3 (training group, active control group, passive control group) mixed analyses of variance (ANOVAs). The composition of every factor is described in the Results section. A solely significant main effect of session or time shows a change from pretest to posttest that is independent of any intervention. On the contrary, a solely significant main effect of group shows differences between groups that are independent of the time of measurement. The former is an indicator for simple retest effects, and the latter demonstrates serious group differences, for instance, based on erroneous randomization. Finally, a significant Group $\times$ Session interaction effect shows an influence of the specific intervention, so this is necessary to demonstrate an effect of training.

RESULTS

Means and standard deviations for every group on the pretest and posttest are shown in Tables 3 and 4. More descriptive

statistics, reliability scores (Table A), the intercorrelations of single tasks (Table B) and constructs (Table C), and means and standard deviations of factor scores (Table D) can be found in the Supporting information. There were no group differences on pretest scores. The results are presented in three parts relating to training effects, near-transfer effects, and far-transfer effects.

Training effects

At first, training gains will be considered. For that purpose, the difference between the first and best training sessions (on average at Session 8; graphs of performance for each working memory capacity task can be found in the Supporting information; Figure A) was calculated for each task and correlated with initial performance revealing rather negative correlations (OS: $r = -.44$, $p = .013$; fRMS: $r = -.25$, $p = .173$; MUN: $r = .11$, $p = .551$; DS: $r = .37$, $p = .043$). Because the performance increase was not steady, we compared the first with the best training session instead

Table 3. Means and standard deviations of working memory, short-term memory, reasoning, processing speed tasks, and cognitive failures

	EG		aCG		pCG	
	Pre	Post	Pre	Post	Pre	Post
RS	3.58 (0.81)	4.13 (0.62)	3.22 (0.86)	3.76 (0.63)	3.35 (0.86)	3.66 (0.82)
OS	4.71 (0.65)	5.18 (0.53)	4.63 (0.44)	4.80 (0.44)	4.57 (0.71)	4.81 (0.59)
DS	2.48 (0.71)	3.24 (0.78)	2.53 (0.67)	2.76 (0.71)	2.53 (0.79)	2.63 (0.79)
MUN	3.99 (0.67)	5.21 (0.62)	3.93 (0.55)	4.59 (0.36)	3.82 (0.77)	4.64 (0.71)
Swaps	11.31 (4.91)	13.50 (5.06)	9.09 (4.53)	11.27 (4.86)	9.72 (4.67)	11.39 (5.83)
nRMS	3.78 (0.61)	4.09 (0.45)	3.67 (0.58)	3.79 (0.49)	3.68 (0.58)	3.52 (0.76)
fRMS	3.72 (0.58)	4.75 (0.54)	3.71 (0.51)	4.10 (0.62)	3.60 (0.65)	4.06 (0.72)
vSTM	5.98 (0.43)	6.05 (0.38)	5.85 (0.50)	5.96 (0.42)	5.83 (0.59)	6.04 (0.44)
nSTM	6.40 (0.43)	6.56 (0.45)	6.35 (0.30)	6.40 (0.37)	6.36 (0.42)	6.42 (0.42)
fSTM	4.30 (0.64)	4.53 (0.64)	4.33 (0.61)	4.50 (0.59)	4.31 (0.70)	4.41 (0.67)
WA	2.65 (1.76)	3.10 (2.18)	1.81 (1.14)	2.35 (1.94)	2.72 (2.10)	2.90 (1.93)
ZN	3.06 (2.39)	2.97 (2.20)	2.35 (1.80)	2.71 (2.32)	2.79 (1.66)	3.66 (2.51)
AN	2.77 (1.96)	3.39 (1.96)	2.26 (1.84)	3.16 (1.86)	2.34 (1.99)	2.34 (1.99)
KW	22.35 (7.03)	25.35 (6.31)	20.68 (5.16)	22.13 (5.16)	22.41 (4.51)	23.76 (4.48)
SI	12.48 (2.95)	13.45 (3.43)	10.71 (3.29)	12.35 (4.02)	12.41 (4.02)	13.90 (3.43)
OE	20.48 (6.87)	21.23 (6.42)	19.48 (5.34)	21.29 (5.15)	20.72 (6.49)	20.62 (5.25)
eKFA	1.13 (0.17)	1.06 ^a (0.14)	1.19 (0.14)	1.14 ^a (0.22)	1.20 (0.28)	1.20 ^a (0.22)
CFQ	2.12 (0.34)	2.12 (0.33)	2.28 (0.28)	2.34 (0.30)	2.26 (0.26)	2.36 (0.26)

Note: $N = 91$. Standard deviations in parentheses.

EG = experimental group; aCG = active control group; pCG = passive control group; RS = reading span; OS = operation span; DS = dot span; MUN = memory updating numerical; nRMS = numerical running memory span; fRMS = figural running memory span; vSTM = verbal short-term memory; nSTM = numerical short-term memory; fSTM = figural short-term memory; WA = verbal reasoning (Berlin Intelligence Structure Test [BIS-4]); ZN = numerical reasoning (BIS-4); AN = figural reasoning (BIS-4); KW = processing speed verbal (BIS-4); SI = processing speed numerical (BIS-4); OE = processing speed figural (BIS-4); eKFA = electronic Questionnaire for Cognitive Failures in Everyday Life; CFQ = Cognitive Failures Questionnaire.

^a $N = 90$.

Table 4. Means and standard deviations of switching reaction times

		EG		aCG		pCG	
		Pre	Post	Pre	Post	Pre	Post
vSW	Single	696.35 (77.05)	660.33 (81.08)	716.55 (53.77)	676.18 (36.73)	718.00 (82.54)	682.82 (70.77)
	Mixed	836.64 (95.68)	759.11 (88.58)	862.72 (66.96)	793.12 (47.73)	845.75 (85.40)	813.38 (75.17)
nSW	Single	798.06 (125.59)	727.69 (131.81)	795.58 (89.19)	751.06 (80.09)	818.16 (113.58)	753.64 (99.57)
	Mixed	854.55 (96.76)	802.87 (140.36)	841.75 (81.55)	795.26 (66.19)	843.80 (94.52)	818.76 (87.15)
fSW	Single	766.71 (103.72)	686.36 (92.73)	794.23 (88.67)	739.46 (91.21)	787.82 (91.47)	744.12 (80.65)
	Mixed	984.82 (123.81)	848.53 (103.56)	1048.79 (153.72)	974.23 (122.14)	967.02 (118.16)	953.61 (123.42)
MST	Single	898.06 (123.41)	718.57 (95.67)	924.43 (95.54)	829.48 (68.14)	917.58 (116.84)	863.99 (105.40)
	Mixed	1233.64 (266.81)	802.44 (126.51)	1256.04 (168.22)	998.30 (66.82)	1225.56 (195.95)	1029.45 (113.73)

Note: $N=91$. Standard deviations in parentheses.

EG = experimental group; aCG = active control group; pCG = passive control group; vSW = verbal switching; nSW = numerical switching; fSW = figural switching; MST = multiple switching task.

of the last session in order to avoid underestimating the training effect. Training gains for working memory tasks are shown in Figure 4.

We investigated training effects based on the standardized (not adaptive) versions of the training tasks applied in the pretest and the posttest. We built a training factor that consisted of the four working memory capacity tasks (OS, DS, MUN, and fRMS). There was a significant group effect, $F(2, 88) = 21.57$, $p = .017$, $\eta^2 = 0.09$, and more importantly, a significant Group $\times$ Session interaction effect, $F(2, 88) = 21.57$, $p < .001$, $\eta^2 = 0.33$. Thus, 9% of the criterion variance could be explained only by belonging to one of the three groups (i.e., regardless of measurement time or intervention between pretest and posttest), and 33% could be explained by specific intervention (i.e., training, quizzes and games, or no contact; Figure 5). Post-hoc tests showed a significant difference between training group and passive control group ($p = .027$) at posttest. There was also a noteworthy difference between training group and active control group, but it was not significant ($p = .067$). The two control groups did not differ.

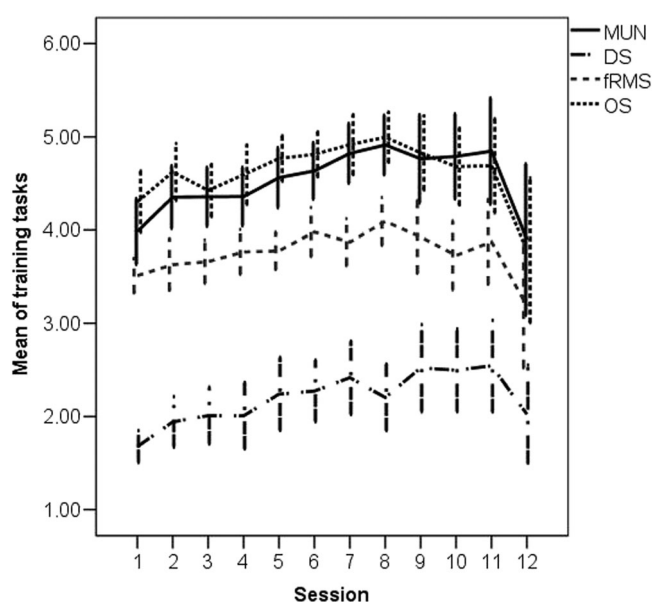


Figure 4. Training gains of working memory tasks. $N=31$. Error bars indicate 95% confidence interval. MUN = memory updating numerical; DS = dot span; fRMS = figural running memory span; OS = operation span

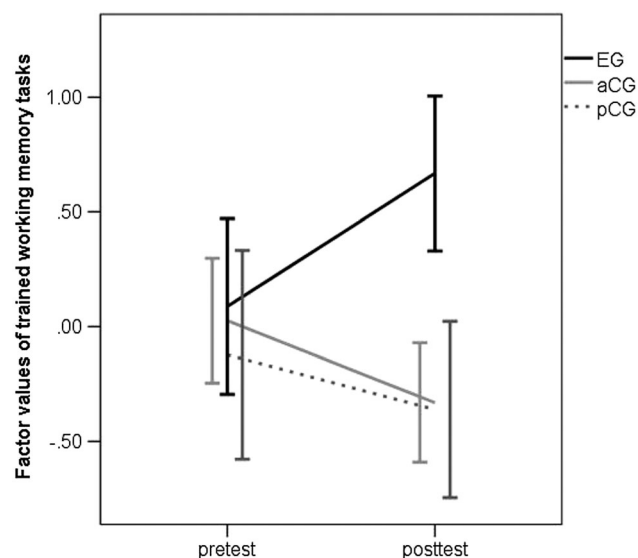


Figure 5. Training effect: interaction plot of working memory training tasks. Error bars indicate 95% confidence interval. $N=91$. EG = experimental group; aCG = active control group; pCG = passive control group

For the training effect of the MST, general switching costs and the switching ratio were considered. Besides large time effects ($\eta^2 = 0.67$ for switching costs and $\eta^2 = 0.69$ for switching ratio), the Group $\times$ Session interactions were significant for switching costs, $F(2, 88) = 5.76$, $p = .004$ ($\eta^2 = 0.12$), and for the switching ratio, $F(2, 88) = 16.05$, $p < .001$ ($\eta^2 = 0.27$), suggesting a successful training of attentional control. Large time effects mean that two-thirds of variance could be explained by the different times of measurement (pretest and posttest). On average, participants in the training group improved from 3.33 to 6.39 during the 12 training sessions, which means they were able to manage six instructions at the end of training.

Near-transfer effects

Next, we examined near transfer to untrained tasks involving the same construct. Three working memory tasks (reading span, swaps, and numerical running memory span) formed a factor as the criterion for a mixed ANOVA. There was again a significant group effect, $F(2, 88) = 3.80$, $p = .026$, $\eta^2 = 0.08$, and the Group $\times$ Session interaction effect was also

significant, $F(2, 88)=4.71$, $p=.011$, $\eta^2=0.10$. Thus, the groups differed from one another but—more importantly—there was a meaningful influence of intervention (Figure 6). In addition, three switching tasks were used for the near transfer of attentional control (verbal memory set switching, numerical response set switching, and figural task set switching). Switching costs, $F(2, 88)=14.84$, $p<.001$, $\eta^2=0.26$, and the switching ratio, $F(2, 88)=4.76$, $p=.011$, $\eta^2=0.10$, showed significant Group $\times$ Session interactions. However, interaction plots in Figure 7 indicate that these interactions are misleading. For working memory and the switching ratio, it can be seen that only the passive control group changed differently in contrast to the experimental group and the active control group. Actually, the interactions disappeared if the passive control group was excluded. Post-hoc tests supported this finding: The training group differed only from the passive control group ($p=.048$). This issue will be discussed later. For switching costs, the interaction plot showed a worsening of the passive control group (higher costs are negative), and again the interaction disappeared without this group.

Far-transfer effects

All Group $\times$ Session interaction effects of ANOVAs for short-term memory, speed, and reasoning were not significant (effect sizes of .01, .02, and .03, respectively), that is, the different interventions (training, quizzes and games, and no contact) showed no influence on these constructs and the hypotheses regarding moderate and far transfer had to be rejected. Complete results can be found in the Supporting information (Table E). Likewise, there was no transfer to the everyday life criteria eKFA ($\eta^2=0.03$) and CFQ ($\eta^2=0.04$). Training gain and transfer on reasoning were not related.

Self-ratings of joy and demand during the study did not differ between the three groups. However, the subjective changes showed a significant group effect: Participants in the experimental group evaluated their abilities as ‘somewhat better than before’, whereas participants in the two control groups estimated their abilities as ‘the same as

before’, $F(2, 83)=3.12$, $p=.049$ (the data of five participants in the passive control group were missing).

DISCUSSION

This study examined the effects of an adaptive and manifold working memory training with an overall duration of 12 hours in a randomized placebo-controlled study design with multiple indicators of several constructs differing in their transfer distance. Although participants in the training group increased their performance on trained tasks, this did not lead to increases in performance on untrained tasks of the same or any other related construct. In fact, the experimental group could not outperform the active control group that played undemanding group games or quizzes. However, both groups differed from the passive control group in working memory capacity and switching gains supporting the assumption that general activation effects—caused by fun, novelty, and breaking out from one’s daily routine—were responsible (note the impact of dopamine level on working memory capacity: Aalto, Brück, Laine, Nägren, & Rinne, 2005; Bäckman & Nyberg, 2013; McNab et al., 2009), but this result could also be an effect of expectation and motivation. Further, these effects failed to appear for short-term memory, speed, reasoning, and everyday life criteria. However, the influence of training may have been overestimated in training studies that did not use an active control group (Au et al., 2014).

As Green et al. (2013) argued, large test–retest effects reduce the possibility of transfer effects because a large amount of learning at pretest precludes much additional improvement through training. Thus, the observed improvements of our active control group might explain the missing transfer effects in the training group. However, we used a passive control group to control for such explanations. Thus, as there were no improvements from pretest to posttest in this passive control group, we can nearly exclude simple test–retest effects in the active control group and training group. This shows that it is important to include passive control groups in training designs.

Our study is in line with the latest studies by Borella et al. (2013a), Stepankova et al. (2013), and Zinke et al. (2013) regarding the results of improvements on trained tasks. Thus, the aging brain is still flexible with regard to both verbal–numerical and figural–spatial working memory capacity as well as switching ability. However, our adaptive multicomponent training did not lead to improvements on similar tasks with different contents (i.e., near-transfer tasks; cf. Shipstead, Hicks, & Engle, 2012b). Neither participants’ age (‘young-olds’ in each study) nor general amount of training (no group in these studies was trained for longer than 12 hours) can explain this difference. As combining different training tasks with different contents and operations was supposed to provide an advantage (as carried out by Zinke et al., 2013), we must question why, by contrast, training with single tasks (e.g., only one updating task, Stepankova et al., 2013; only one complex span task, Borella et al., 2013a) produced near-transfer effects. One explanation might be the larger amount of time spent with one single task, which

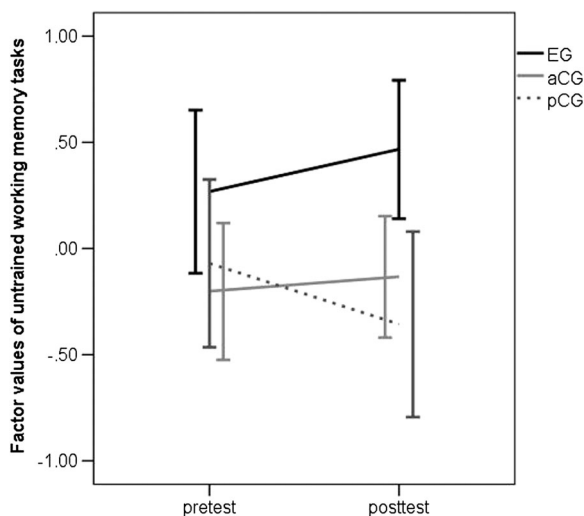


Figure 6. Near-transfer effect: interaction plot of working memory transfer tasks. Error bars indicate 95% confidence interval. $N=91$. EG = experimental group; aCG = active control group; pCG = passive control group

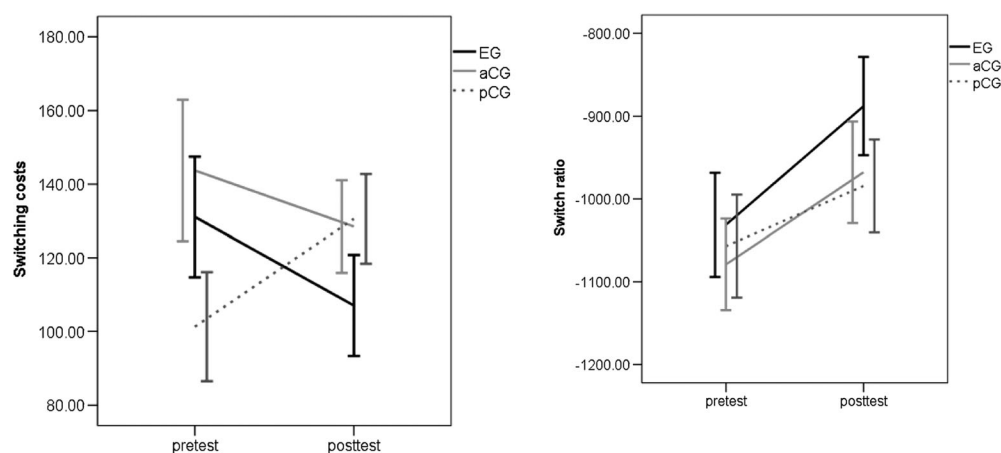


Figure 7. Near-transfer effect: interaction plots of switching transfer tasks (left: costs; right: ratio). Error bars indicate 95% confidence interval. $N=91$. EG = experimental group; aCG = active control group; pCG = passive control group

likely enabled participants to acquire some strategies that were somehow transferable to other tasks. Regarding absolute training times, our participants spent 10 minutes on average per training task in each session, and this added up to 2 hours across the entire training. Thus, although our participants had a higher dose of training all in all (12 hours), they spent less time on every single training task than participants in other studies. This shows one big disadvantage of variable training in contrast to single-task training: If training sessions are kept to a tolerable overall length in order to not bore or overstrain participants, the amount of time spent on every task is reduced. Additionally, participants need some time to adapt to each new task, which also reduces the effective training time. Therefore, multitask trainings need larger training times, and future studies should consider this issue.

Consequently, as we supposed near transfer to be a requirement for far transfer, we found no transfer to fluid intelligence, although there were good reasons to expect it: (i) Working memory and fluid intelligence share substantial variance owing to similar demands and restrictions (e.g., the ability to simultaneously maintain information units); (ii) different studies have already reported that they found this transfer (e.g., Jaeggi et al., 2008); and (iii) our training procedure should support transfer (e.g., adaptive manifold sessions with adequate distance). However, our finding of no far transfer to reasoning is in line with that of other studies (e.g., Chooi & Thompson, 2012; Harrison et al., 2013; Redick et al., 2013; Richmond et al., 2011; Thompson et al., 2013). We must question why working memory is assumed to be improvable but reasoning is not, although working memory shares about 72% of its variance with reasoning ability (Oberauer, Schulze, Wilhelm, & Süß, 2005; but note the lower values of Ackerman, Beier, & Boyle, 2005, whose meta-analysis resulted in an overlap of 23–41%, depending on computation base). Is training tapping only the remaining 28% and the shared variance is unaffected? More likely, the improvement does not occur at the construct level of working memory, as mentioned in the previous paragraph.

Stephenson and Halpern (2013) proposed that only task-content-specific overlaps between the n-back task and Raven's matrices (both used primarily in studies with positive transfer) have been responsible for previous findings. Likewise, von Bastian and Oberauer (2013b) showed that

transfer after working memory training is specific—however, they reported transfer effects to reasoning. Sprenger et al. (2013) found only process-specific instead of general transfer as well. Nevertheless, a final answer cannot be given yet, and as there are some design issues (see next section), our data do not warrant the conclusion that transfer is not possible. In fact, there is first evidence that transfer to reasoning is not only possible, but that it even remains 2 years after intervention—at least in younger adults (Schmiedek, Lövdén, & Lindenberger, 2014). There are already promising approaches other than working memory training to enhance older adults' cognition (e.g., Ball et al., 2011; Kramer & Willis, 2002; Park, Gutches, Meade, & Stine-Morrow, 2007), and there are also hints that a combination of different demands within a training task is more promising than a simple combination of different tasks as we applied. For example, video games have shown positive effects on multitasking performance (Anguera et al., 2013; Basak, Boot, Voss, & Kramer, 2008; Strobach, Frensch, Soutschek, & Schubert, 2012b), but transfer effects are still under discussion regarding, for instance, differential effects of motivation and expectations between training and control groups (e.g., Boot, Blakely, & Simons, 2011; Boot & Simons, 2012; Schubert & Strobach, 2012).

Limitations

There are further methodological issues that could account for the missing near and far transfer. One main limitation is that our training regime has never been implemented in younger populations, so—in fact—there is no empirical evidence for its efficiency as of yet. Regarding the four categories mentioned in the introduction, the training was ineffective in older adults, so it is uncertain whether the training itself or the aging cohort is responsible for the zero effects. Additionally, we used a novel transfer measure (eKFA) that was probably not suited to detect transfer. Although it is a promising and reliable instrument, its validity is not completely proven (Lange & Süß, 2014).

The group size ($n=30$) may have been too small to detect effects, although other studies with positive transfer have employed even a shorter training time and fewer participants per group. For example, Sprenger et al. (2013) used 20 hours

of training and a group size between 57 and 70 without evidence of general improvements after working memory training.

Furthermore, the adaptive algorithm was not the best possible in the present study: Participants began every session at the lowest level of difficulty instead of adjusting it to reflect their personal gains from the last session. However, as described in the Method section, we increased the starting level and decreased the number of examples after Session 4, so we tried to adapt the training to participants' enhanced performance during the course of the training. More importantly, two tasks with different contents and different operations (MUN and DS) had a performance-based stop criterion, so the rather low starting points did not affect the possibility of reaching personal high scores in these tasks. Overall, the adaptivity of our training is improvable, but we do not consider this constraint as responsible for the zero effects.

To sum up, the lack of near transfer to working memory is a novel finding, but owing to these methodological issues, we cannot conclude that near transfer is impossible. However, even an improved performance on working memory tasks does not inevitably mean that there is an improvement at the construct level, as Harrison et al. (2013) noted. Future studies should pay more attention to a careful examination of near transfer, that is, training should maximize the possibility of transfer (e.g., manifold, adaptive, and sufficient length) and study design should minimize misinterpretations (e.g., active and passive control group, multiple indicators for each criterion, and analysis on construct level).

Future directions

First of all, the training should be applied to younger adults in order to test its effectiveness. Further, this study should be repeated with a larger sample and a longer training period (particularly with regard to the training time of single tasks), and, additionally, the adaptive algorithm could be optimized to rule out alternative explanations. However, it should be considered that—besides the effectiveness—the efficiency of training is also an important feature. Very small effects, which can be detected only by large samples, might not justify the effort of such an extensive study. Perhaps criteria other than cognitive tests should be used to evaluate the effectiveness of working memory training. We tried to examine a transfer effect to everyday life via the eKFA, an innovative ambulatory assessment of cognitive failures (a new questionnaire was applied via mobile phones four times daily), but further indicators of everyday competence are needed (cf. Richmond et al., 2011; Willis & Schaie, 1986). Otherwise, if pure activation is sufficient to produce improvements, training should not generally be regarded as an intervention but as a supplement to a healthy lifestyle like a balanced diet or exercise. Maybe there are just different and exchangeable ways of enhancing the mind, and perhaps specific programs—like working memory trainings—are necessary only for people who do not have access to other options. Finally, there are training benefits that are independent of cognitive effects, namely, increased life satisfaction, which might in turn positively affect the mental, cognitive, and physical health of elderly people. Those indirect effects should be given more attention in future studies.

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